# POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Probability and Statistics | Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

1. a) In a survey, it was found that 64 families bought milk in litre. A frequency distribution is given as 8

| Milk in litres | Frequency |
|---|---|
| 5 - 9 | 7 |
| 10 - 14 | 10 |
| 15 - 19 | 13 |
| 20 - 24 | 18 |
| 25 - 29 | 8 |
| 30 - 34 | 5 |
| 35 - 39 | 3 |

Construct the histogram and frequency polygon of the distribution.

b) The average electricity consumption and standard deviation of two sets of households are as follows. 7

| | Set I | Set II |
|---|---|---|
| Mean | 65 units | 72 units |
| Standard deviation | 13 units | 18 units |
| Number of households | 50 | 40 |

i. Which set is more variable?

ii. Calculate the coefficient of variation of the combined set.

2. a) From a group of 3 Nepalese 4 Indians and 5 Americans a committee of 4 persons are selected at randomly. Find the probability that the committee consists 8

i. 3 Nepalese and 1 Indian

ii. 1 Nepalese, 1 Indian and American



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iii. 4 Americans

b) In a certain factory machines, I, II, and III are all producing springs of the same length of their production machine I, II and III produces 1%,2%,3% defective springs respectively. Of the total production the machine I, II and III produces 35%,25% and 40% respectively. if one spring is selected at random from total springs produce in a day, find 7

i. The probability that it is defective

ii. The conditional probability that it was produced by machine III

3. a) Define Mathematical Expectation. An unbiased coin is tossed until a head appears. What is the expectation of number of tosses required to get first head? 8

b) Define Standard Normal distribution. In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution 7

4. a) The number of accidents in a year attributed to taxi driver in a city is Poisson distribution with mean 3. out of 1000 taxi driver find the number of taxi drivers 7

i. More than 3 accidents in a year.

ii. Less the 2 accidents in a year.

iii. No accidents in a year.

b) Define joint and marginal probability density functions. Two random variables X and Y have the following probability density function. 8

$$f(x,y)=\begin{cases}8xy & 0<x<y<1\\ 0 & otherwise\end{cases}$$

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Find:

i. Marginal probability density functions of x and Y

ii. Conditional functions of X on y Y on X

5. a) Define Interval estimation. A random sample of 100 articles selected from a batch of 2000 articles shows that the average diameter of the article is 0.354 with standard deviation 0.048. Find 95% confidence limits for the average diameter of this batch of 2000 articles. 7

b) In a random sample of 400 students of the university teaching departments, it was found that 300 students failed in the examination. In another random sample of 500 students of the affiliated colleges, the number of failures in the same examination was found to be 300. Find 8



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out whether the proportion of failures in the university teaching departments is significantly greater than that of affiliated colleges. Use $\alpha = 0.05$.

6. a) An I.Q. test was administered to 5 persons before and after they were trained.The resultis given below: 7

| Candidates | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| IQ before Training | 110 | 120 | 123 | 132 | 125 |
| IQ after training | 120 | 118 | 125 | 136 | 121 |

Is the training effective at 5% level of significance?

b) The following are marks obtained by 9 students in Mathematics (X) and Statistics (Y) : 8

| X | 40 | 65 | 60 | 25 | 85 | 35 | 45 | 70 | 80 |
|---|---|---|---|---|---|---|---|---|---|
| Y | 30 | 85 | 65 | 35 | 90 | 35 | 75 | 75 | 45 |

i. Develop the estimating regression equation of X on Y.
ii. Find the correlation coefficient
iii. Find Coefficient of determination.

7. Write short notes on: (**Any two**) 2×5

a) Properties of a good Estimator.
b) Error in hypothesis testing.
c) Methods of primary data collection.

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